Data Science and Artificial Intelligence

Machine Learning

Multi layer Perceptron

Lecture No. 2

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Recap of Previous Lecture

- Topic (Perceptions)
- Topic
- Topic
- Topic
- Topic

Topics to be Covered

- Topic
- Topic
- Topic
- Topic
- Topic

Questions

Perceptron → MLP (NN)

"Sometimes later becomes never.
Do it now." –Unknown

REVISION

Perceptron ⇒

- Start with w = all zero
- $y_i(w^T x_i) > 0 \Rightarrow$ Correct

  $y_i(w^T x_i) \leq 0 \Rightarrow w^{new} = w^{old} + \eta y_i x_i$

REVISION

Perceptron

- If the data is non linearly seperable then this will not work
- algorithm will never stop.

REVISION

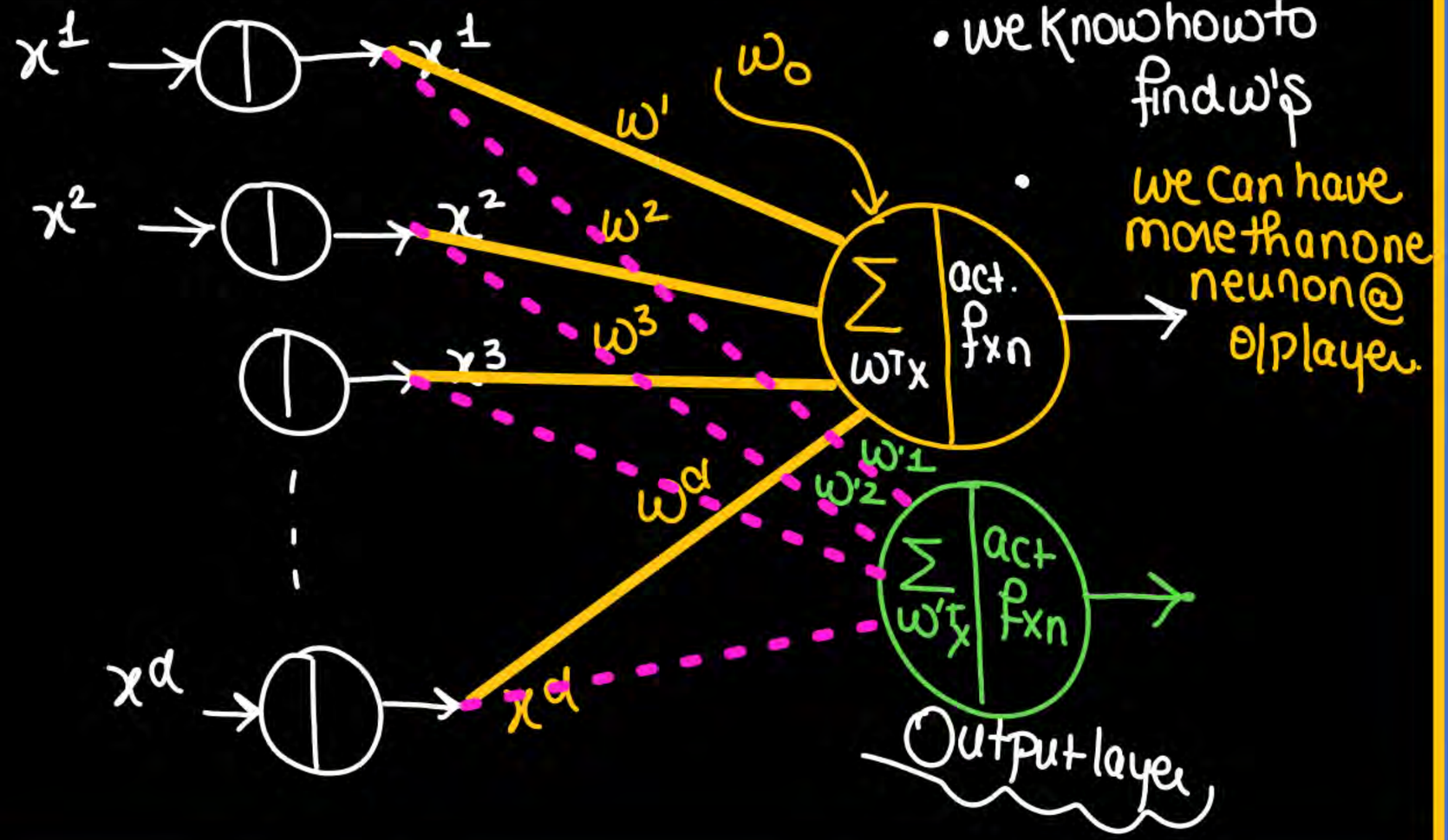
Perceptron Jo Kal Padhatha

→ Single layer Perceptron

↳ 2 layer Input layer / Output layer

Basics of Machine Learning

Input layer → neurons ⇒ Simply Storage



- we know how to find w's
- we can have more than one neuron @ o/p layer.

REVISION

- The problem that exist is this setup can work only with the linearly seperable data

#Q. You are a Hollywood producer. You have a script in your hand and you want to make a movie. Before starting, however, you want to predict if the film you want to make will rake in huge profits, or utterly fail at the box office. You hire two critics A and B to read the script and rate it on a scale of 1 to 5 (assume only integer scores). Each critic reads it independently and announces their verdict. Of course, the critics might be biased and/or not perfect, therefore you may not be able to simply average their scores. Instead, you decide to use a perceptron to classify your data. There are three features: a constant bias, and the two reviewer scores. Thus $f_0 = 1$ (a constant bias), f_1 = score given by reviewer A, and f_2 = score given by reviewer B.

3 features

f_0	f_1	f_2
1	Critic A	Critic B

f_0	f_1	f_2
1	50 option	50 option

25 points in data.

f_0	f_1	f_2
1	1	1
1	1	2
1	1	3
1	1	4
1	1	5
1	2	1
	2	2
	2	3
	2	4
	2	5

25 points in data

#Q. Suppose that, instead of learning whether or not the movie is profitable, you want to learn a perceptron that will always output hat $\hat{Y} = 1$ when the total of the two reviewer scores is more than 8, and hat $\hat{Y} = -1$ otherwise. Is this possible? If so, what arethe weights ω_0, ω_1, and ω_2 that will make this possible?

So algo ⇒

$f_1 + f_2 > 8 \Rightarrow$ Class 1

$f_1 + f_2 < 8 \Rightarrow$ Class -1

Classifier already given

$\Rightarrow \begin{cases} \omega_1 = 1 \\ \omega_2 = 1 \\ \omega_0 = -8 \end{cases}$

$f_1 + f_2 - 8 > 0$ class 1

$f_1 + f_2 - 8 < 0$ Class -1

$(\omega_1 f_1 + \omega_2 f_2 + \omega_0 f_0) > 0$ Class 1

$\omega_1 f_1 + \omega_2 f_2 + \omega_0 f_0 < 0$ Class 0

#Q. Instead of either part (a) or part (b), suppose you want to learn a perceptron that will always output hat $\hat{Y} = +1$ when the two reviewers agree (when their scores are exactly the same), and will output hat $\hat{Y} = -1$ otherwise. Is this possible? If so, what are the weights ω_0, ω_1, and ω_2 that will make this possible?

No.

- This is not linearly Seperable Case
- So algo fail.

#Q. Train the perceptron to generate hat $\hat{Y} = 1$ if the movie returns a profit, hat $\hat{Y} = -1$ otherwise. The initial weights are $\omega_0 = -1$, $\omega_1 = 0$, $\omega_2 = 0$ Present each row of the table as a training token, and update the perceptron weights before moving on to the next row. Use a learning rate of $\alpha = 1$ After each of the training examples has been presented once (one epoch), what are the weights?

Movie Name	A	B	Profit	
Pellet Power	1	1	No	→ -1
Ghosts!	3	2	Yes	→ 1
Pac is bac	4	5	No	→ -1
Not a Pizza	3	4	Yes	→ 1
Endless Maze	2	3	Yes	→ 1

- $\omega = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}$

1) $y_i(\omega^T x_i) = -1(-1+0+0) = 1$

$y_i(\omega_0 + \omega_1 A + \omega_2 B) \Rightarrow -1(-1) = 1$ ✓

2) $y_i(\omega^T x_i) \Rightarrow 1(-1+0+0) = -1$

$\omega^{new} = \omega^{old} + \alpha y_i x_i$

$$w^{new} = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \\ 2 \end{bmatrix}$$

3) $y_i(w^T x_i) = -1(0 + 3\times4 + 2\times5) = -ve$

$$w^{new} = \begin{bmatrix} 0 \\ 3 \\ 2 \end{bmatrix} + \underbrace{\alpha}_{\to 1}\, \underbrace{y_i}_{\to -1}\, x_i$$

$$= \begin{bmatrix} 0 \\ 3 \\ 2 \end{bmatrix} - \begin{bmatrix} 1 \\ 4 \\ 5 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \\ -3 \end{bmatrix}$$

4. $y_i(w^T x_i) = \underset{-ve}{1(-1 - 1\times3 - 3\times4)}$

$$w^{new} = w^{old} + \alpha y_i x_i$$

$$= \begin{bmatrix} -1 \\ -1 \\ -3 \end{bmatrix} + \begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix}$$

5. $y_i(w^T x_i) = \underset{+ve}{1(0 + 2\times2 + 1\times3)}$

$$w = \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix} \Rightarrow \begin{array}{l} w_0 = 0 \\ w_1 = 2 \\ w_2 = 1 \end{array}$$

#Q. Consider a supervised learning problem with N examples where each example is a point in d-dimensional space. Imagine we apply the perceptron learning algorithm on the data and it converges (finds a perceptron that separates the data) after t iterations.

What is the runtime of this algorithm as a function of N, d, and t ?

→ Time Complexity

~~a) Nat~~

~~b) (N+N)(d+1)t~~

c) N(d+1)t → we need bias term w has d+1 parameters.

d) none

#Q. Given a feature space with 2 Boolean features < x1, x2 >.

How many Boolean functions are representable by a perceptron?

x_1	x_2	Ylabel
0	0	0/1
0	1	0/1
1	0	0/1
1	1	0/1

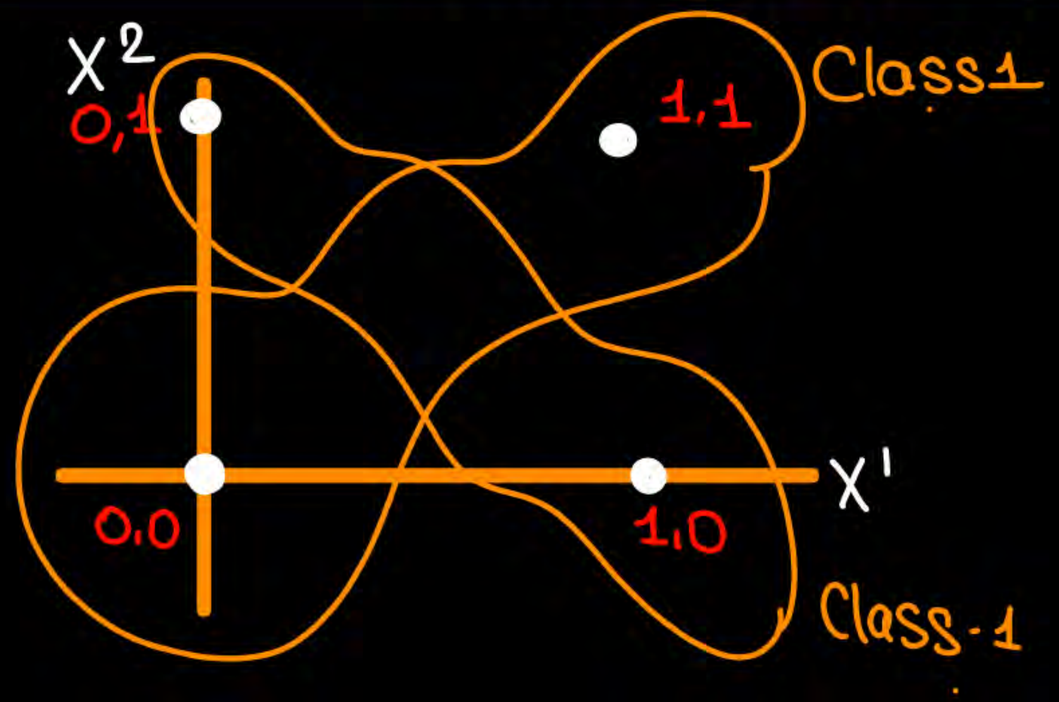
4 spaces to fill

⇒ So we can get 16 combination of Ylabels.

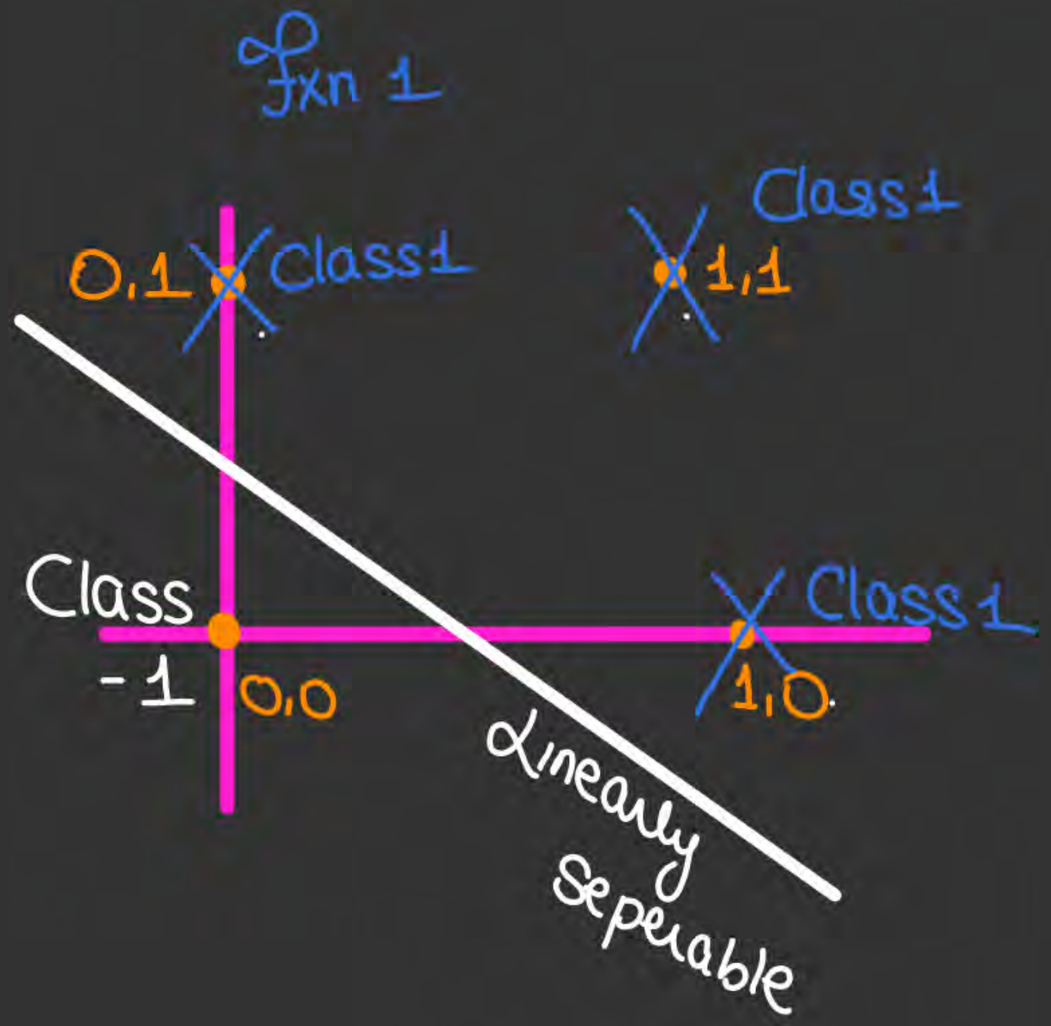
⇒ 16 functions that we can create

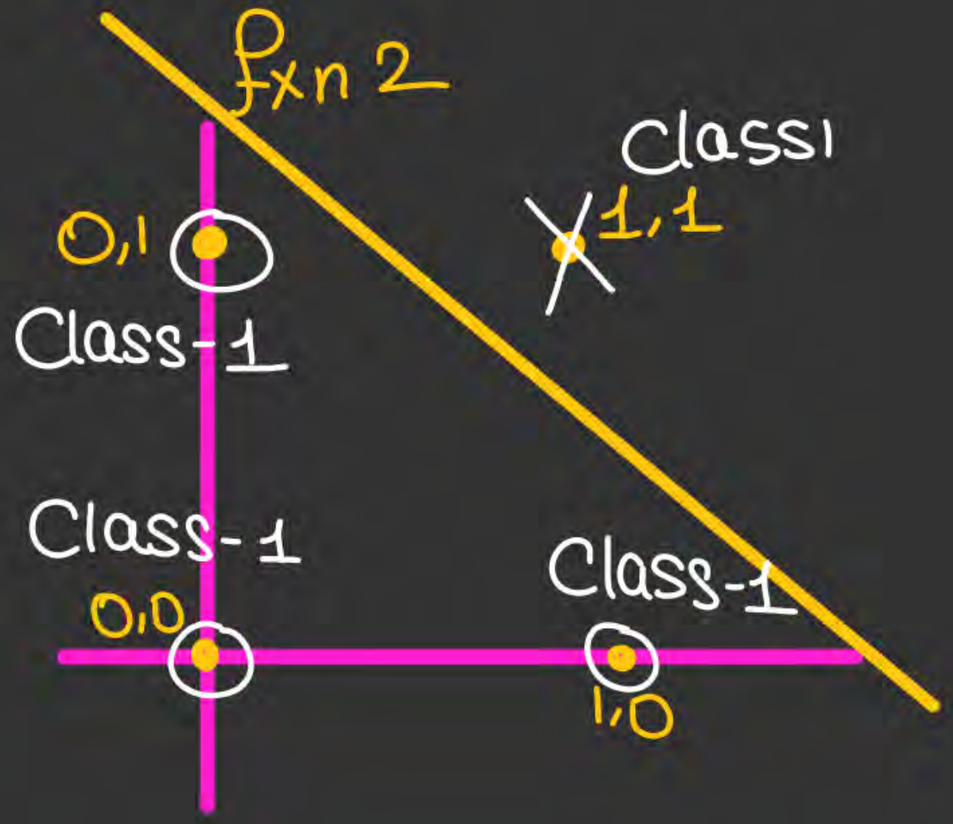
Perceptron learning

Problem in Single Layer Perceptron...



XOR, XNOR are not linearly seperable


fxn 1
0,1
Class1
Class1
1,1
Class
-1
0,0
Class1
1,0
Linearly
Seperable

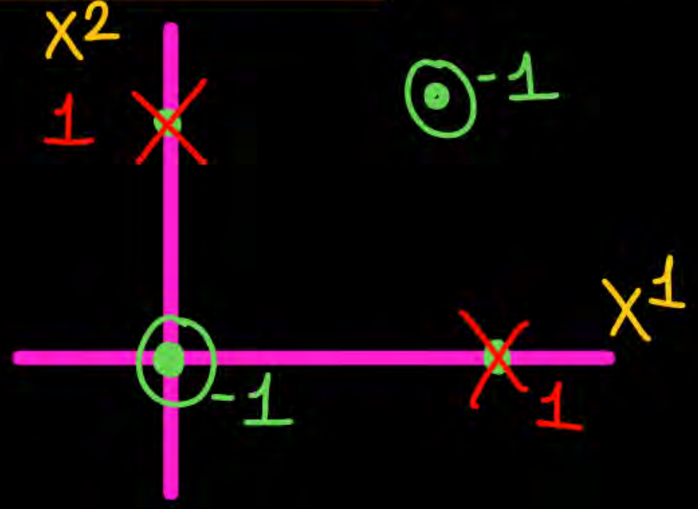

fxn 2
Class1
1,1
0,1
Class-1
Class-1
0,0
Class-1
1,0

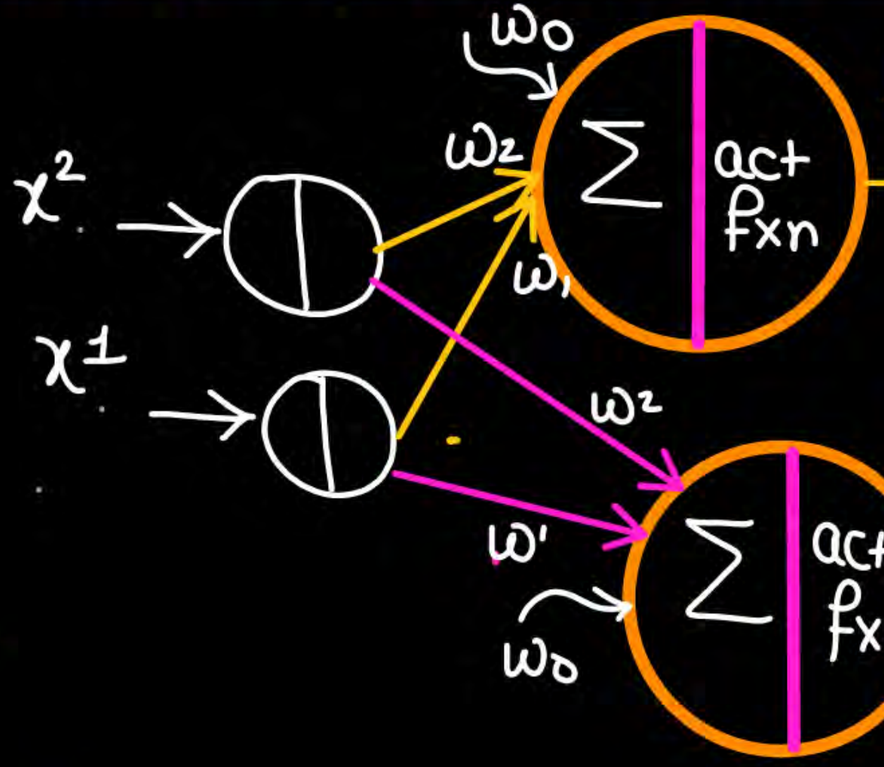
Perceptron learning

algo fails @ Non linearly seperable data

Problem in Single Layer Perceptron...

x^2	x^1	Y
0	0	-1
0	1	1
1	0	1
1	1	-1





function Y=1 if any X_1 or X_2=1

function Y=1 only when Both X_1, X_2 = 1

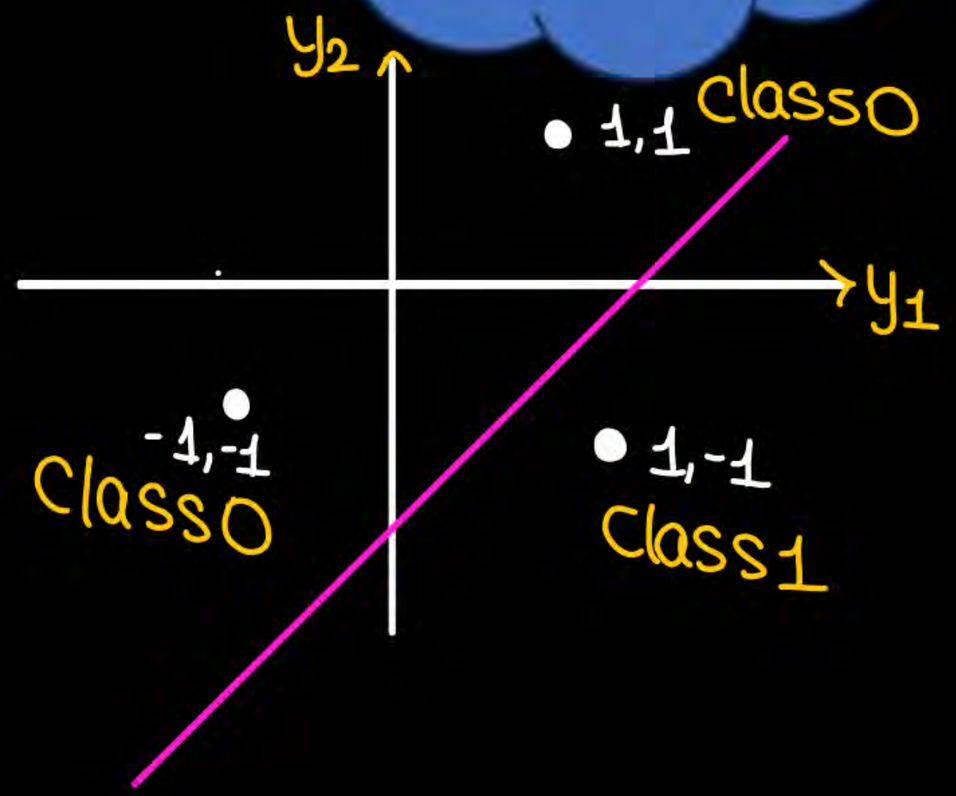
Perceptron learning

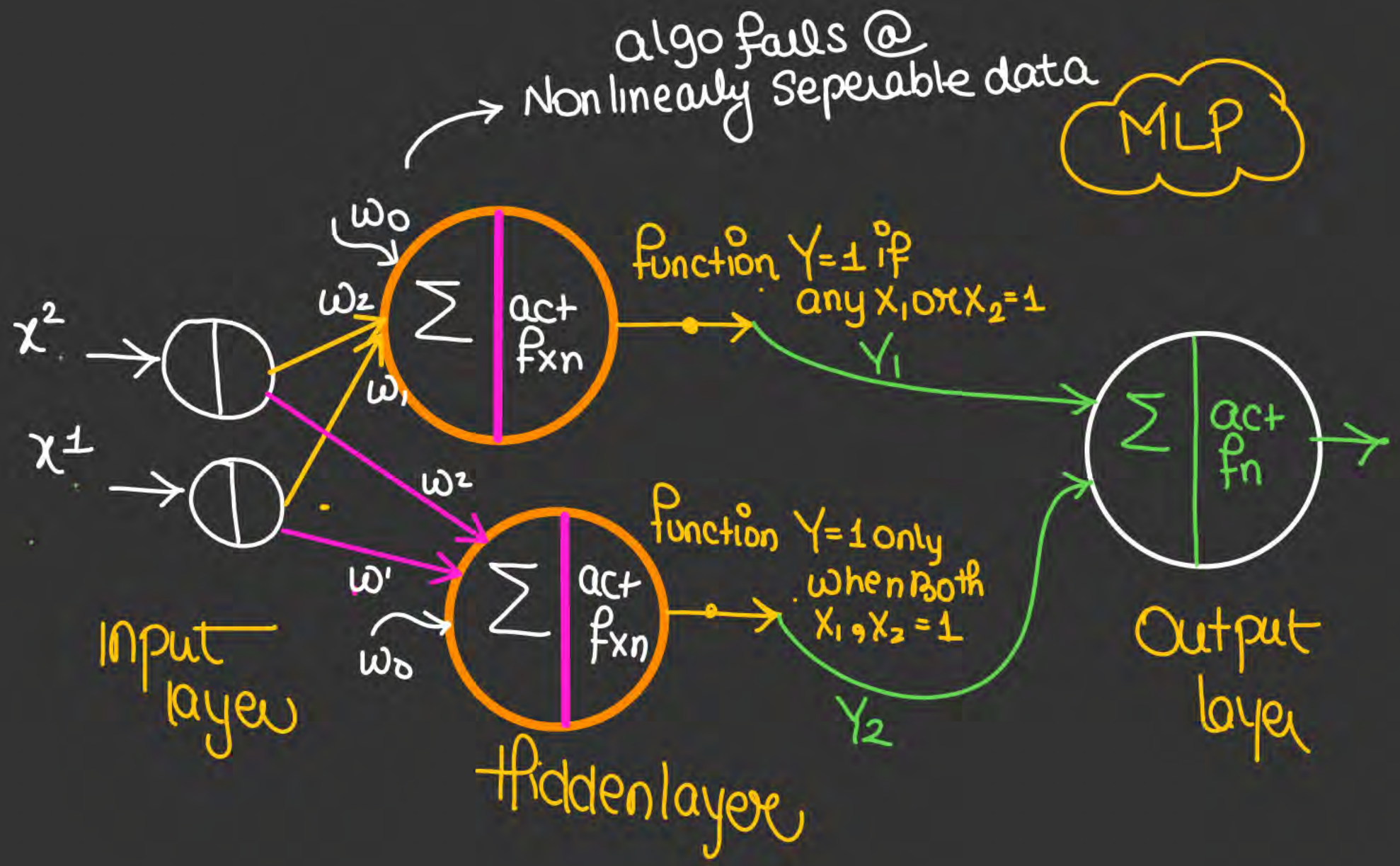
Non linear transformation

x^2	x^1	Y_1 (fxn1)	Y_2 (fxn2)	Y label (actual)
0	0	-1	-1	0
0	1	1	-1	1
1	0	1	-1	1
1	1	1	1	0

How to find the function for this transformation

...




algo fails @
Non linearly seperable data
MLP
x^2
x^1
w_0
w_2
w_1
w^2
w'
w_0
Σ
act fxn
function Y=1 if
any X_1 or X_2=1
Y_1
function Y=1 only
when Both
$X_1, X_2 = 1$
Y_2
Input layer
Hidden layer
Output layer

Perceptron learning

Non linear transformation

Approach : Find the parameters of the function during the training ...

So • by transforming the non linearly seperable data into higher dimension or by simply transformation we can get linearly seperable data

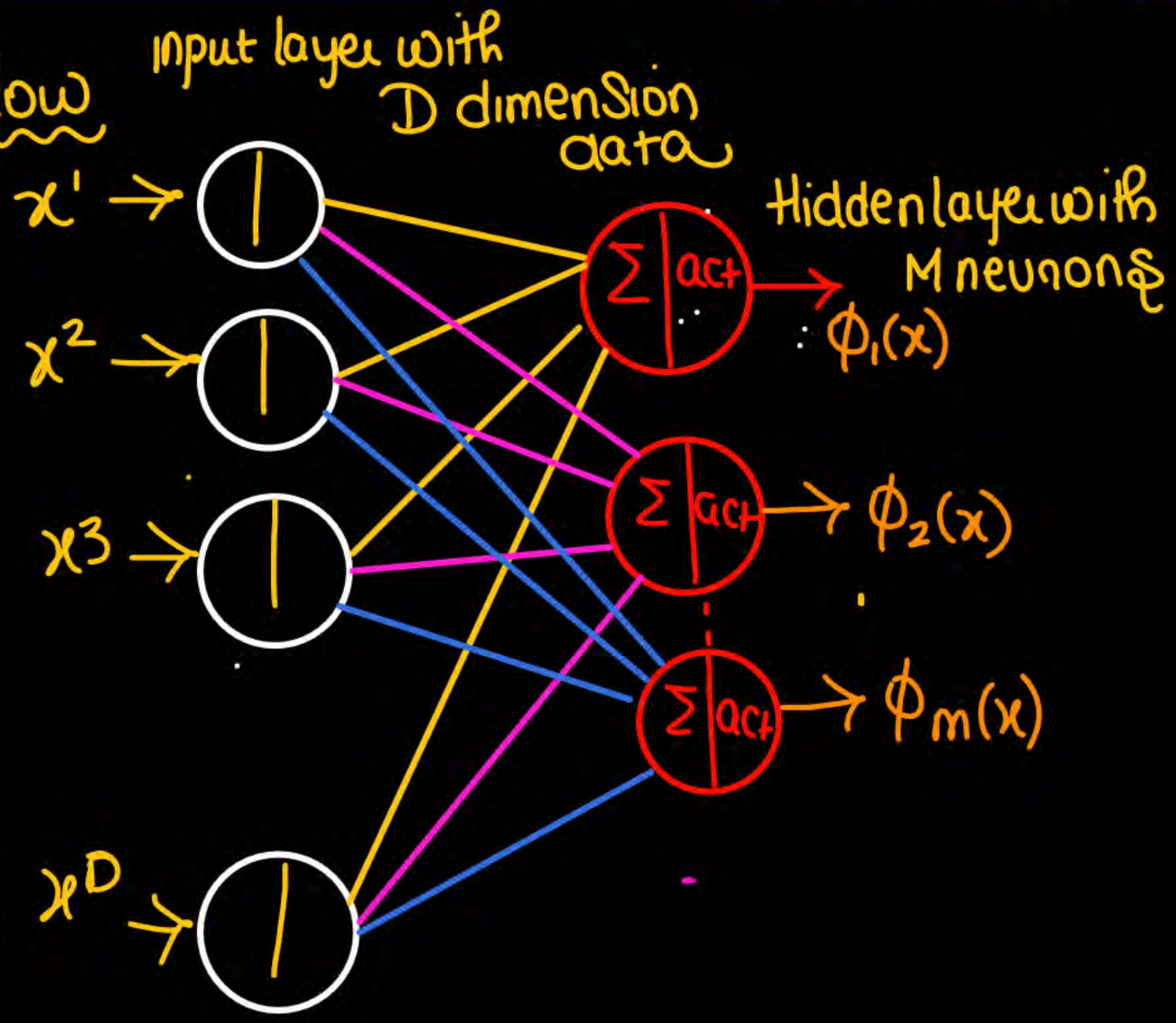
• the MLP has input layer
output layer
and hidden layers,

• we can have multiple hidden layers also.

Perceptron learning

Non linear transformation

Now



Example of EXOR

So D dimension data → M dimen data

new dimension

$$\begin{bmatrix} \phi_1(x) \\ \phi_2(x) \\ \vdots \\ \phi_m(x) \end{bmatrix}$$

Perceptron learning

Non linear transformation

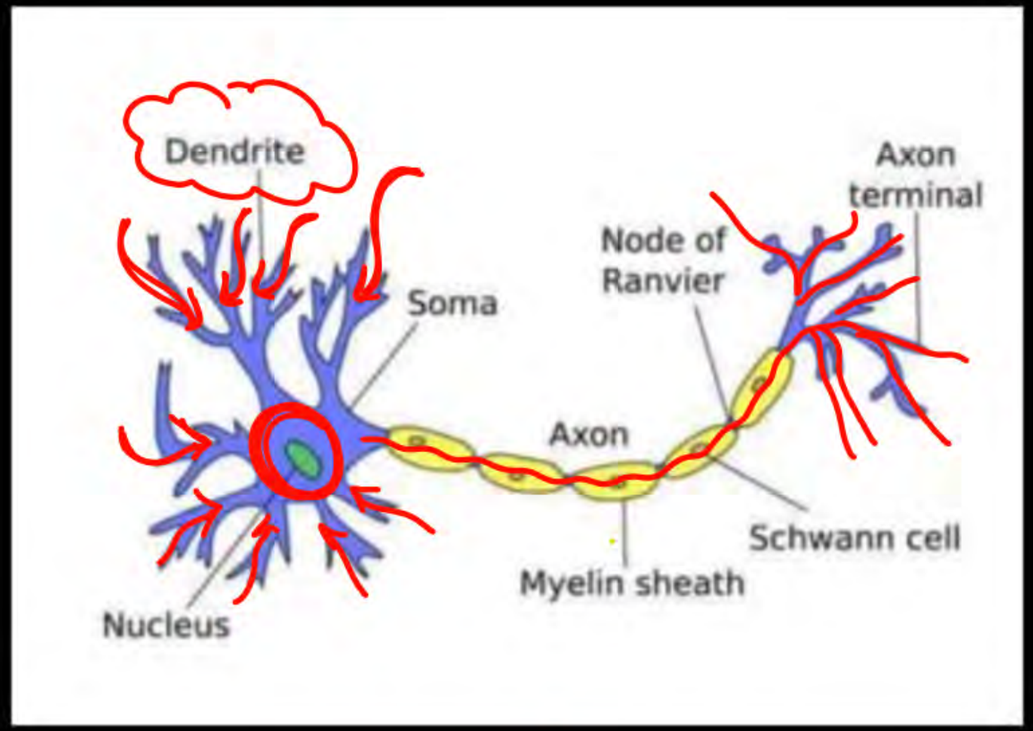
- So (x_i) of D dimension $\longrightarrow \phi(x_i) \Rightarrow \begin{bmatrix} \phi_1(x_i) \\ \vdots \\ \phi_m(x_i) \end{bmatrix}$

Example of EXOR

Perceptron learning

Neural Networks

Neuron work same as the single cell learnt yesterday…

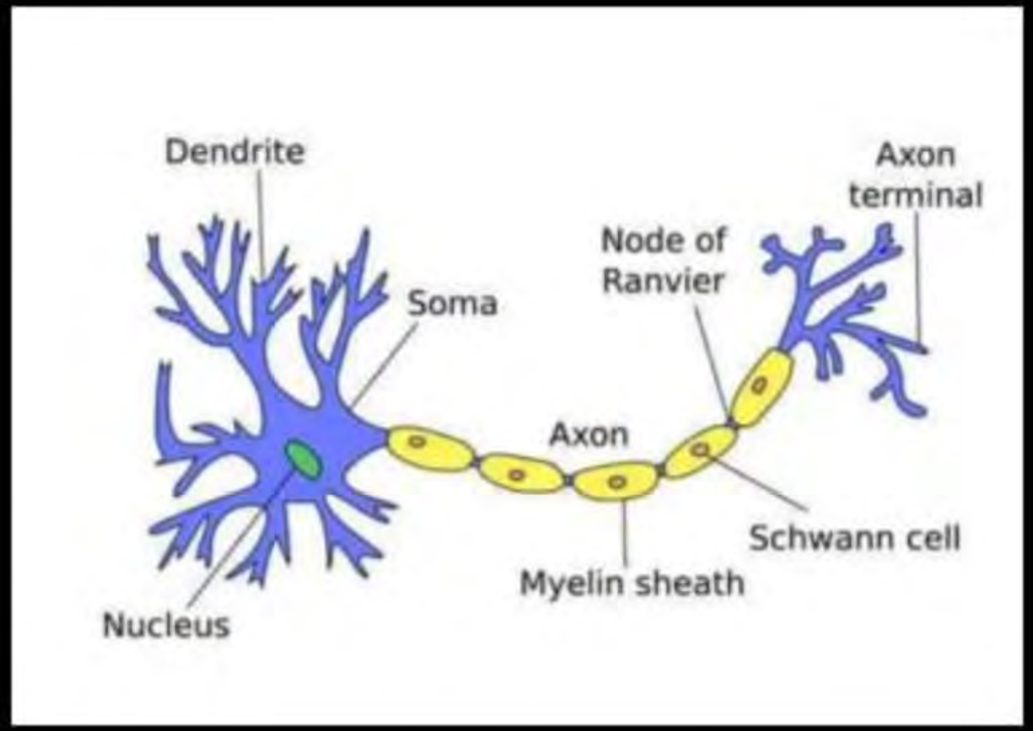


- In our Brain these neurons are interconnected
  Similar to connection seen in previous slide
- So MLP is a part of NN.

Perceptron learning

Neural Networks

How weights are assigned in the neurons …



Single Vs Multi Layer Perceptrons

Single Layer Perceptron has just two layers of input and output. It only has single layer hence the name single layer perceptron.

Perceptron learning

Single Vs Multi Layer Perceptrons

- Limitations of the Perceptron

- Linear Separability: The perceptron can only solve problems where the data is linearly separable. If the data cannot be separated by a straight line (or hyperplane in higher dimensions), the perceptron will not converge.
- Non-linear Problems: For problems like the XOR problem, a single-layer perceptron fails as it cannot capture the non-linearity.

Perceptron learning

Single Vs Multi Layer Perceptrons

A multilayer perceptron is a type of feed-forward artificial neural network that generates a set of outputs from a set of inputs. An MLP is a neural network connecting multiple layers in a directed graph, which means that the signal path through the nodes only goes one way. The MLP network consists of input, output, and hidden layers. Each hidden layer consists of numerous perceptron's which are called hidden layers or hidden unit.

Single Vs Multi Layer Perceptrons

Multilayer Perceptron (MLP):

- Consists of multiple layers of neurons.
- Can solve non-linear problems using hidden layers and non-linear activation functions.
- Trained using backpropagation.

Multi Layer Perceptrons

What the hidden layer does ??

Multi Layer Perceptrons

How can we generate a transformation function from R (D) to R(M)

Multi Layer Perceptrons

So the linear classifier in the new dimension space seems to be non linear in lower/original dimension space

MLP to NN

In NN we need that the activation function must be :

1.

2.

MLP to NN

- **Multilayer Perceptron (MLP):**
- An MLP is a class of feedforward artificial neural network (ANN) composed of multiple layers of neurons.
- It includes an input layer, one or more hidden layers, and an output layer.
- Each neuron in one layer is connected to every neuron in the next layer, making it a fully connected network.
- **Neural Network (NN):**
- The term "neural network" is a broad term that encompasses a variety of architectures, including MLPs, recurrent neural networks (RNNs), and more.
- Neural networks are computational models inspired by the human brain, capable of learning complex patterns through layers of interconnected neurons.

MLP to NN

The term "neural network" is a broad term that encompasses a variety of architectures, including MLPs, convolutional neural networks (CNNs), recurrent neural networks (RNNs), and more.

Feed Forward NN

What is a Feedforward Neural Network?

- A Feedforward Neural Network (FNN) is a type of artificial neural network where the connections between nodes do not form a cycle.
- It is the simplest form of artificial neural network, consisting of an input layer, one or more hidden layers, and an output layer.
- Information moves in one direction—from the input layer, through the hidden layers, to the output layer.

Feed Forward NN

- Architecture

- Input Layer: Consists of input neurons that receive the initial data.
- Hidden Layers: Consist of neurons that process inputs from the previous layer using weights, biases, and activation functions. Can be multiple hidden layers, each adding more complexity and capacity to the network.
- Output Layer: Consists of output neurons that produce the final prediction or classification.

Perceptron learning

Feed Forward NN

- We can decide on number of hidder layers, no of nodes in a layer and activation functions.

Feed Forward NN

- Number of parameters in the Feed Forward NN

THANK - YOU